Handoff for the Vendor: GOAT SMTP

Oct 13, 2026 · @Melvin Wilson Jr.

TLDR

IBCM has set up a mailbox, goat@ibcm.com.au, for your service to send email through Microsoft 365. Configured 13 Oct 2026. It uses OAuth 2.0 client credentials, so there is no mailbox password.

- **You receive:** a tenant ID, client ID and the SMTP settings in this document. The client secret comes separately, through a secure channel.
- **You do:** configure your service, send one test message to an external address, and reply with the results listed under "What to send back".
- **Watch for:** the From address must be goat@ibcm.com.au. Use AUTH XOAUTH2 on smtp.office365.com, port 587, with STARTTLS. Tokens last about an hour. Replies to goat@ are rejected by design.
- **Expect:** your first attempt may fail with 535 5.7.3 while permissions take effect. Retry after a few hours and tell us the result either way.
- **Never email** the client secret or an access token.

Where each value goes

Part of your setup	Setting	Value
OAuth (token request)	Tenant ID	0299386a-a135-4d14-9d36-b571b6e3a637
OAuth (token request)	Client ID	da247c5e-5598-4a2d-9391-7277f282a828
OAuth (token request)	Client secret	Sent separately. It starts with h8_8Q, so you can check you have the right one
OAuth (token request)	Scope	https://outlook.office365.com/.default
OAuth (token request)	Grant type	client_credentials
SMTP	Server, port, encryption	smtp.office365.com, 587, STARTTLS

Part of your setup	Setting	Value
SMTP	Login method	OAuth 2.0 (XOAUTH2), not a password
SMTP	Username and From address	goat@ibcm.com.au

The client secret belongs only in the OAuth settings. It is never the SMTP password. If your product's SMTP screen offers only a username and password, tell us before going further: it needs OAuth 2.0 support for SMTP.

Part 1: Implement, test and report back

Work through the steps in order. Stop at the first failure and send the feedback listed at the end. You do not need to finish first.

Settings

Setting	Value
Authentication	OAuth 2.0 client credentials. No mailbox password
Tenant ID	0299386a-a135-4d14-9d36-b571b6e3a637
Client ID	da247c5e-5598-4a2d-9391-7277f282a828
Client secret	Sent separately. Not in this document
Token endpoint	https://login.microsoftonline.com/0299386a-a135-4d14-9d36-b571b6e3a637/oauth2/v2.0/token
Scope	https://outlook.office365.com/.default
SMTP server	smtp.office365.com
Port and encryption	587 with STARTTLS (required)
SMTP login method	AUTH XOAUTH2
SMTP username	goat@ibcm.com.au
From address	goat@ibcm.com.au. Send from this address only

Step 1: Get an access token

```
curl -X POST "https://login.microsoftonline.com/0299386a-a135-4d14-9d36-
b571b6e3a637/oauth2/v2.0/token" \
  -H "Content-Type: application/x-www-form-urlencoded" \
  -d "client_id=da247c5e-5598-4a2d-9391-7277f282a828" \
  -d "client_secret=<CLIENT_SECRET>" \
  -d "scope=https://outlook.office365.com/.default" \
  -d "grant_type=client_credentials"
```

Expected: HTTP 200 and a JSON body with a `token_type` of Bearer, an `expires_in` of about an hour, and an `access_token`. There is no refresh token. Request a new token before the old one expires.

Step 2: Check the token

Decode the token on your own machine, not on a website. Its `roles` claim should contain `SMTP.SendAsApp`. If it does not, stop and tell us.

Step 3: Log in over SMTP

Connect to smtp.office365.com on port 587, upgrade with STARTTLS, then authenticate. The XOAUTH2 string is the base64 encoding of this, where 0x01 is the control character with that code:

`user=goat@ibcm.com.au` 0x01 `auth=Bearer <access_token>` 0x01 0x01

A healthy session looks like this, with the token redacted:

```
S: 220 ... Microsoft ESMTP MAIL Service ready
C: EHLO your.host.name
S: 250-... 250-STARTTLS ...
C: STARTTLS
S: 220 2.0.0 SMTP server ready
   (TLS handshake, then EHLO again)
S: 250-AUTH LOGIN XOAUTH2 ...
C: AUTH XOAUTH2 <base64 string>
S: 235 2.7.0 Authentication successful
```

A 535 5.7.3 reply means authentication failed. See "Common errors" at the end.

Step 4: Send a test message

Send one message from goat@ibcm.com.au to an external address you control. Optionally, reply to it and tell us what you see: the reply should bounce with an explanation. A message addressed to goat@ibcm.com.au itself is rejected by design, so do not use it as a

recipient.

What to send back

Item	What we need
Timeline	Date, time and time zone of each attempt
Token result	HTTP status, expires_in, and whether `roles` contains SMTP.SendAsApp. Never the token itself
SMTP log	The full session with every response code. Redact the token and the base64 string
Result	Success, or the exact error text, copied and not paraphrased
Screenshot 1	Your service's email or SMTP configuration screen, with the secret masked
Screenshot 2	The send result or log entry in your service
Screenshot 3	The received message in the external mailbox, ideally with full headers
Your configuration	Product and version, login method selected, server, port, encryption setting, From address, and any library used
Optional	What the bounce looked like, if you replied to the test message

Send this to whoever sent you this document. If a step failed, send what you have at that point.

Implementation notes

- Use AUTH XOAUTH2, not LOGIN or PLAIN with a password. There is no mailbox password to use.
- Cache the token and request a new one a few minutes before it expires.
- Keep the secret in a secrets manager. Do not log it or the token.
- Some SMTP libraries, such as MailKit, have a built-in XOAUTH2 mechanism. Use it rather than building the string by hand.

Part 2: What was configured and how it works

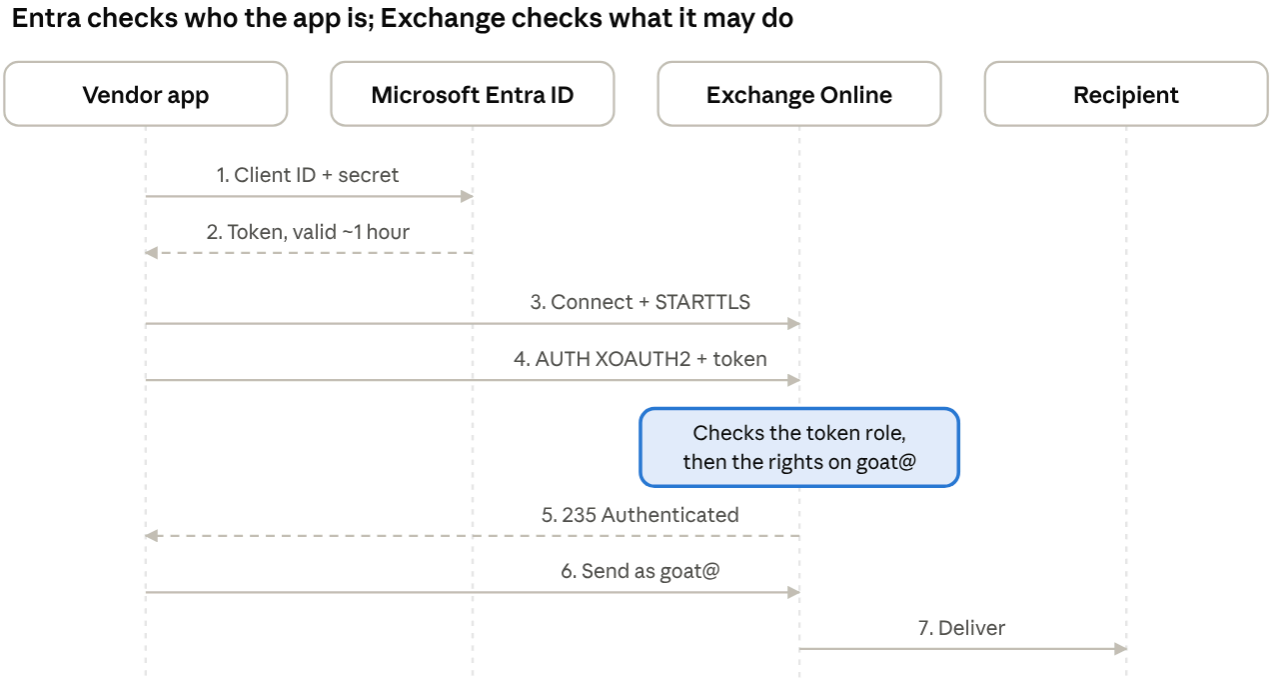
This section explains the design behind the settings in Part 1, so you can reason about failures and report them precisely.

The short version

Your service proves its identity to Microsoft Entra ID with a client ID and secret, and receives a short-lived access token. It presents that token to Exchange Online over SMTP. Exchange accepts the login only for the one mailbox, goat@ibcm.com.au, that IBCM granted the app access to.

The pieces

Piece	What it is	Why it exists
Shared mailbox goat@ibcm.com.au	A mailbox with no password and no license	Gives your service a sending identity that nobody can sign in to
Entra app registration "GOAT - Vendor SMTP"	Your service's identity in IBCM's directory, identified by the client ID	Holds the secret and the permission that makes OAuth sending possible
Client secret	A password for the app, not for the mailbox	Proves to Entra that the caller is your service. It expires
API permission SMTP.SendAsApp	A role granted to the app, with administrator consent	Puts the SMTP.SendAsApp role in the token, which tells Exchange the app may use SMTP
Exchange service principal	The app's identity as Exchange sees it	Lets Exchange attach permissions to the app
FullAccess and SendAs on goat@	Two permissions on one mailbox	Decide which mailbox the app may use, and let it send as that address
SMTP AUTH on goat@	A per-mailbox switch, enabled on goat@ while the tenant default is off	Keeps the SMTP login path closed by default, and open where it is needed
Reply bounce rule	A mail flow rule on goat@	Rejects replies, because nobody reads this mailbox



sequence · one send, 4 parties, 7 steps

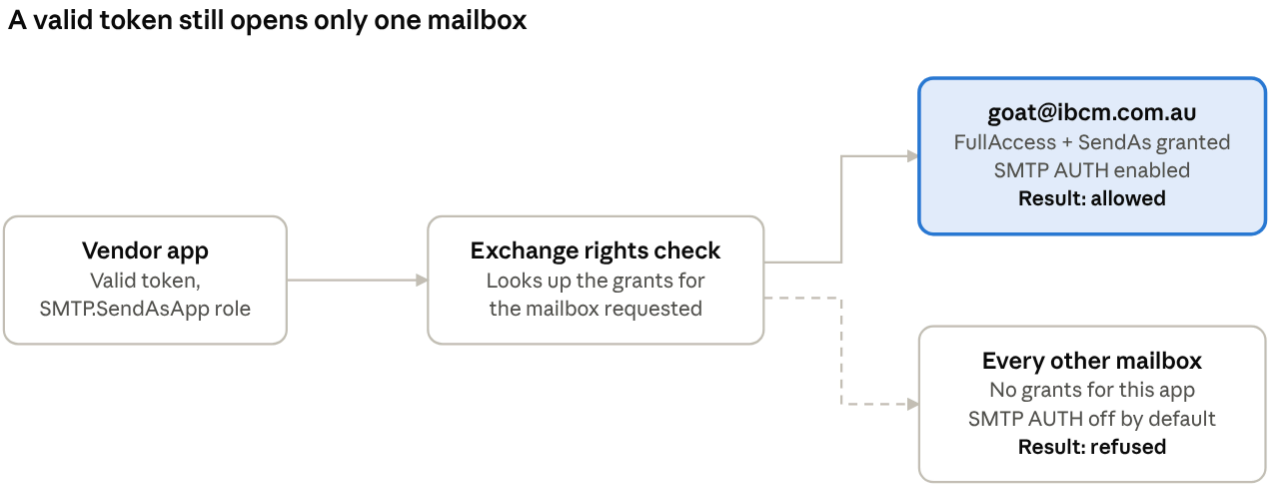
Dashed arrows are replies from the server. Steps 1 and 2 get the token. Steps 3 to 7 send the message.

What happens on every send

1. Your service asks Entra for a token with its client ID and secret, using the client credentials grant and the scope `https://outlook.office365.com/.default`.
2. Entra checks the secret. If it is valid, Entra returns an access token that lasts about an hour. The token carries the `SMTP.SendAsApp` role.
3. Your service connects to smtp.office365.com on port 587 and upgrades to TLS with STARTTLS.
4. It authenticates with AUTH XOAUTH2, giving goat@ibcm.com.au as the user and the token as the credential.
5. Exchange validates the token, matches it to the app's service principal, and checks that the app has rights on goat@ and that SMTP AUTH is enabled there.
6. If those pass, Exchange replies 235 and your service sends the message with goat@ibcm.com.au as the sender.
7. Exchange delivers the message to the recipient.

Two checks happen before any message leaves: Entra checks who the app is, and Exchange checks what it may do.

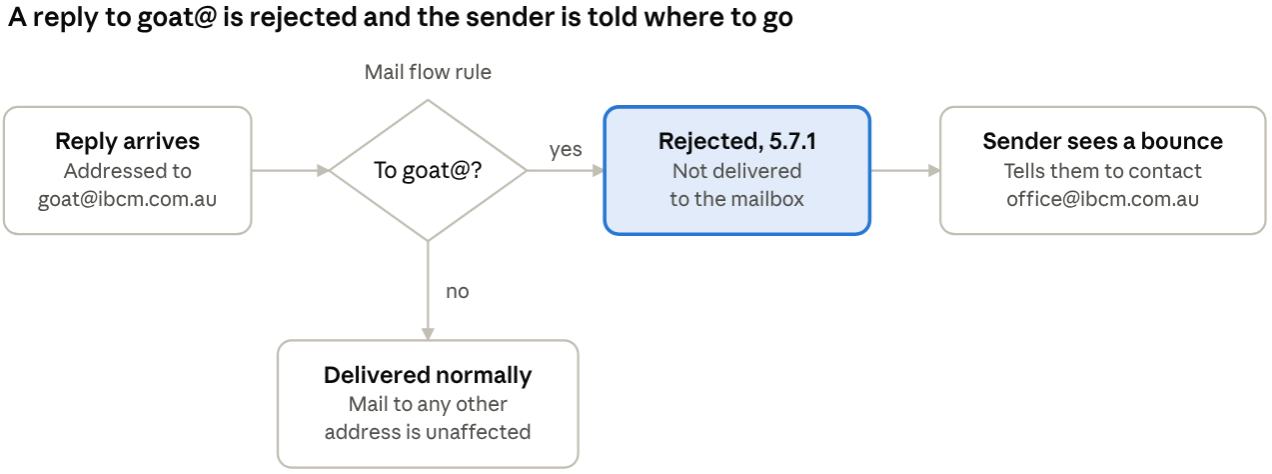
Why the app can only send as goat@



access boundary · 1 mailbox allowed, the rest refused

A valid token proves identity and carries the SMTP role. It does not name a mailbox. The mailbox comes from the grants IBCM created in Exchange: FullAccess and SendAs on goat@ only. The app has no grants on any other mailbox, and SMTP AUTH is off by default across the tenant. If your service tried to log in as a different mailbox it would be refused, though IBCM has not tested that case.

What happens to replies



reply flow · 1 decision, 2 outcomes

goat@ is a send-only identity. A mail flow rule rejects anything addressed to it, and the sender's mail server shows them a bounce reading "This mailbox is not monitored. Please contact office@ibcm.com.au."

What this means for you:

- Replies to your emails will not reach your service. If people need to reply, set a Reply-To header to an address that someone monitors.
- Delivery-failure notices addressed to goat@ are rejected too. Do not rely on them. Check the SMTP response codes at send time and handle failures there.
- Mail to any other IBCM address is unaffected.

Security and limits

- Treat the secret like a password: keep it in a secrets manager, and never put it in email, tickets or logs.
- No mailbox password exists, so nobody can sign in to goat@ interactively.
- The secret has an expiry date. Plan for rotation: your service should be able to load a new secret without a code change.
- Approximate sending limits from Microsoft's published SMTP AUTH guidance: about 30 messages per minute and 10,000 recipients per day per mailbox, with few concurrent connections. IBCM has not tested these. This setup is not for bulk sending.
- Not configured: receiving mail, IMAP, POP and the Microsoft Graph API. Ask if you need any of them.

What has not been tested

IBCM finished the configuration on 13 Oct 2026. No message has been sent through it yet. Your test is the first end-to-end check, which is why the feedback list in Part 1 is detailed.

Common errors

Copy the exact text of any error you see when you report back. These are the likely meanings.

What you see	Likely meaning	What to do
Token request fails with AADSTS7000215	The secret is wrong. A common cause is copying the secret's ID instead of its value	Check the value you were given. Tell us if it still fails
Token request fails with AADSTS7000222	The secret has expired	Tell us. IBCM issues a new secret

What you see	Likely meaning	What to do
Token is issued but `roles` lacks SMTP.SendAsApp	The permission is not reaching the token	Stop and tell us. Do not retry the SMTP login
SMTP 535 5.7.3 Authentication unsuccessful	Most often one of: permissions still taking effect, a wrong username, a malformed XOAUTH2 string, or the wrong token scope	Retry after a few hours. Check the username is goat@ibcm.com.au, the scope is `https://outlook.office365.com/.default`, and the string uses real 0x01 characters. Then tell us
SMTP 535 5.7.139, or text saying SMTP client authentication is disabled	SMTP AUTH is off for the mailbox or the tenant	Tell us. IBCM will check the mailbox setting
A "not permitted to send as this sender" style rejection	The From address is not goat@ibcm.com.au	Send from goat@ibcm.com.au only
550 5.7.1 on a recipient of goat@ibcm.com.au	The reply bounce rule, working as designed	Use an external recipient for tests
Cannot connect	Outbound port 587 is blocked on your network	Use port 587 with STARTTLS. Check your firewall

If a test fails, send what you have. A partial report is more useful than waiting.